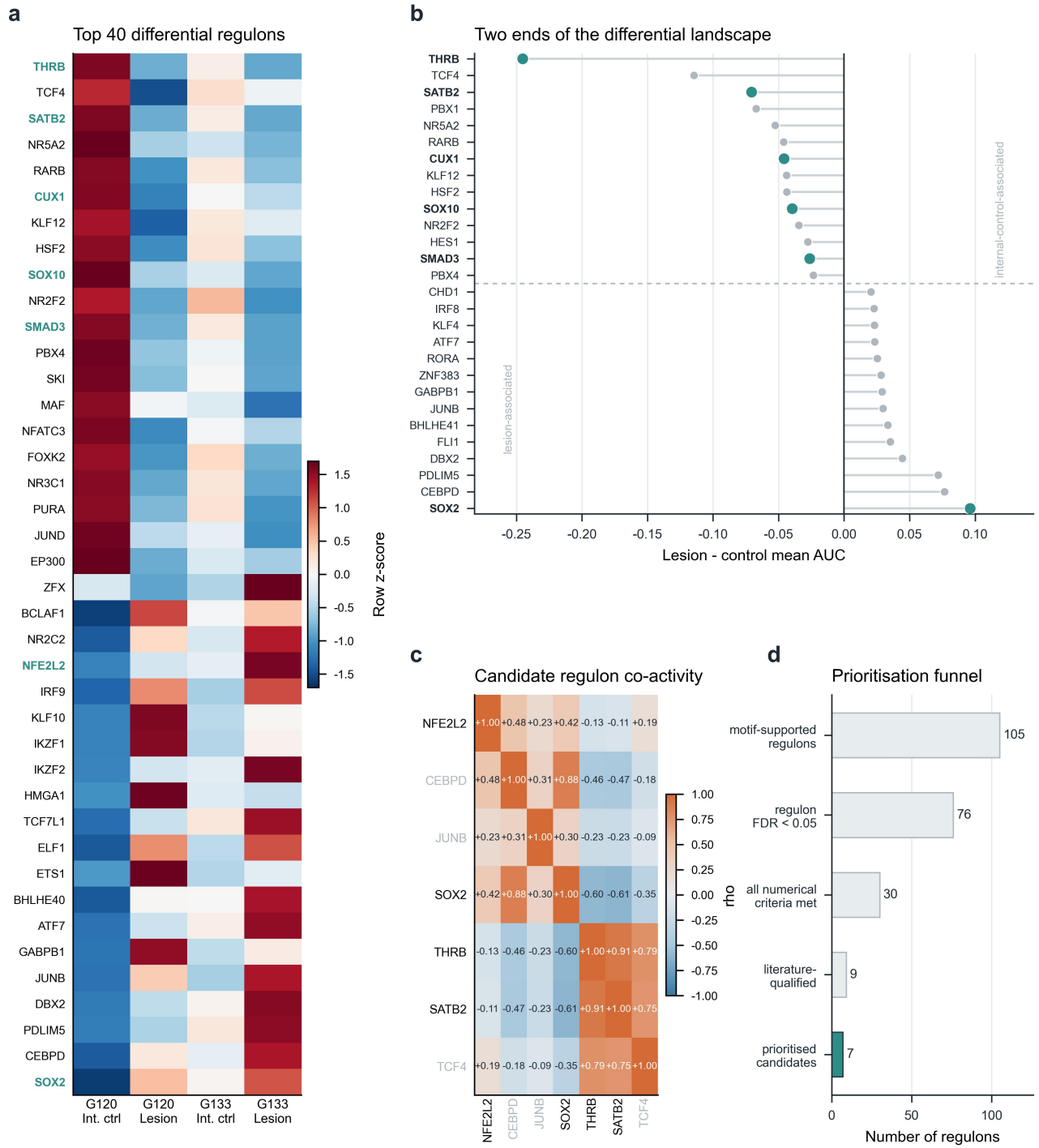

a
Top 40 differential regulons
THRB
TCF4
SATB2
NR5A2
RARB
CUX1
KLF12
HSF2
SOX10
NR2F2
SMAD3
PBX4
SKI
MAF
NFATC3
FOXK2
NR3C1
PURA
JUND
EP300
ZFX
BCLAF1
NR2C2
NFE2L2
IRF9
KLF10
IKZF1
IKZF2
HMGA1
TCF7L1
ELF1
ETS1
BHLHE40
ATF7
GABPB1
JUNB
DBX2
PDLIM5
CEBPD
SOX2
G120 Int. ctrl
G120 Lesion
G133 Int. ctrl
G133 Lesion
Row z-score
1.5
1.0
0.5
0.0
-0.5
-1.0
-1.5
b
Two ends of the differential landscape
lesion-associated
internal-control-associated
-0.25
-0.20
-0.15
-0.10
-0.05
0.00
0.05
0.10
Lesion - control mean AUC
c
Candidate regulon co-activity
NFE2L2 +1.00 +0.48 +0.23 +0.42 -0.13 -0.11 +0.19
CEBPD +0.48 +1.00 +0.31 +0.88 -0.46 -0.47 -0.18
JUNB +0.23 +0.31 +1.00 +0.30 -0.23 -0.23 -0.09
SOX2 +0.42 +0.88 +0.30 +1.00 -0.60 -0.61 -0.35
THRB -0.13 -0.46 -0.23 -0.60 +1.00 +0.91 +0.79
SATB2 -0.11 -0.47 -0.23 -0.61 +0.91 +1.00 +0.75
TCF4 +0.19 -0.18 -0.09 -0.35 +0.79 +0.75 +1.00
NFE2L2
CEBPD
JUNB
SOX2
THRB
SATB2
TCF4
rho
1.00
0.75
0.50
0.25
0.00
-0.25
-0.50
-0.75
-1.00
d
Prioritisation funnel
motif-supported regulons
105
regulon FDR < 0.05
76
all numerical criteria met
30
literature-qualified
9
prioritised candidates
7
0
50
100
Number of regulons